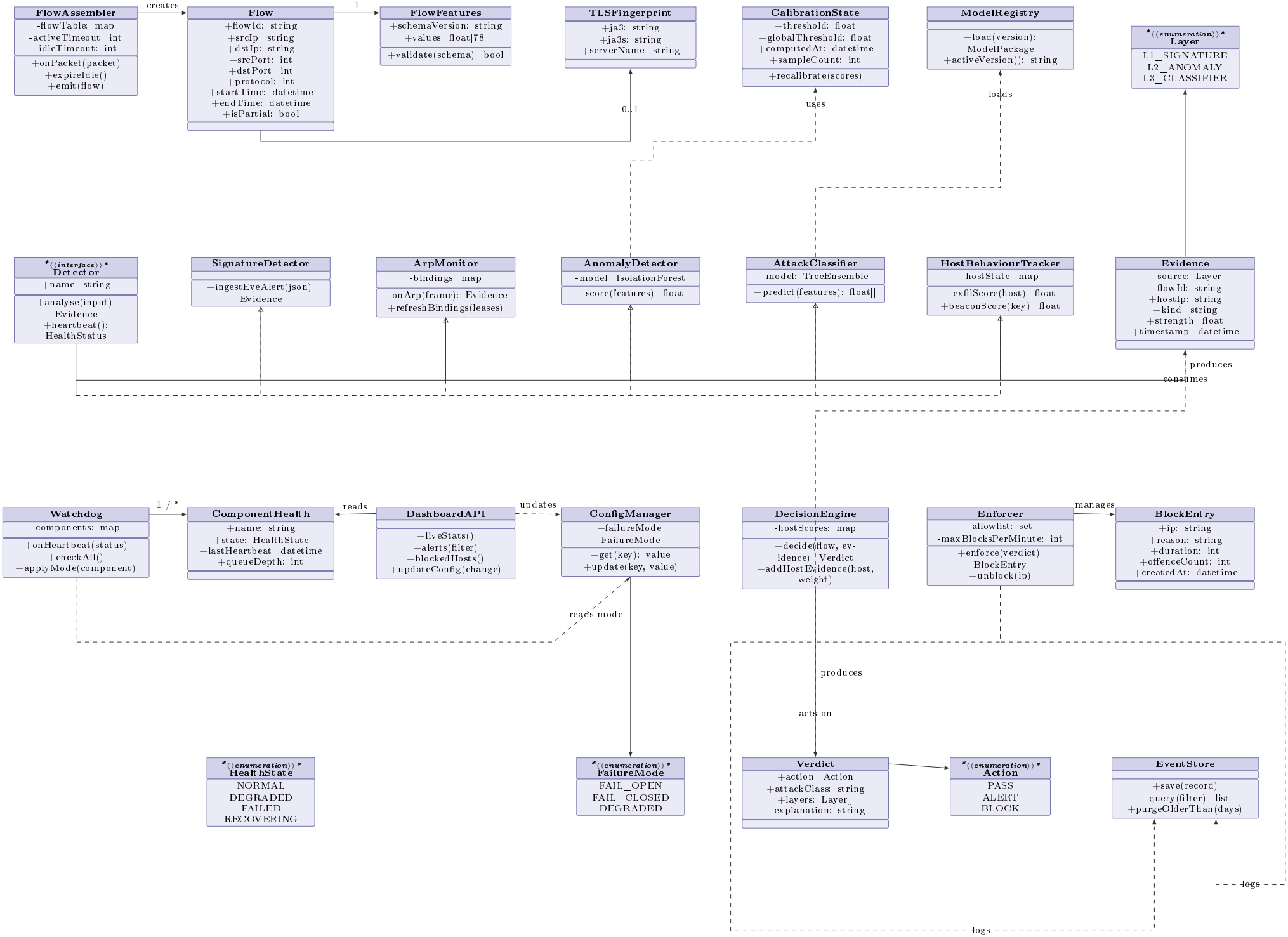

FlowAssembler
-flowTable: map
-activeTimeout: int
-idleTimeout: int
+onPacket(packet)
+expireIdle()
+emit(flow)
creates
Flow
+flowId: string
+srcIp: string
+dstIp: string
+srcPort: int
+dstPort: int
+protocol: int
+startTime: datetime
+endTime: datetime
+isPartial: bool
1
FlowFeatures
+schemaVersion: string
+values: float[78]
+validate(schema): bool
TLSFingerprint
+ja3: string
+ja3s: string
+serverName: string
0..1
CalibrationState
+threshold: float
+globalThreshold: float
+computedAt: datetime
+sampleCount: int
+recalibrate(scores)
uses
ModelRegistry
+load(version): ModelPackage
+activeVersion(): string
loads
«enumeration» Layer
L1_SIGNATURE
L2_ANOMALY
L3_CLASSIFIER
«interface» Detector
+name: string
+analyse(input): Evidence
+heartbeat(): HealthStatus
SignatureDetector
+ingestEveAlert(json): Evidence
ArpMonitor
-bindings: map
+onArp(frame): Evidence
+refreshBindings(leases)
AnomalyDetector
-model: IsolationForest
+score(features): float
AttackClassifier
-model: TreeEnsemble
+predict(features): float[]
HostBehaviourTracker
-hostState: map
+exfilScore(host): float
+beaconScore(key): float
Evidence
+source: Layer
+flowId: string
+hostIp: string
+kind: string
+strength: float
+timestamp: datetime
produces
consumes
Watchdog
-components: map
+onHeartbeat(status)
+checkAll()
+applyMode(component)
1 / *
ComponentHealth
+name: string
+state: HealthState
+lastHeartbeat: datetime
+queueDepth: int
reads
DashboardAPI
+liveStats()
+alerts(filter)
+blockedHosts()
+updateConfig(change)
updates
ConfigManager
+failureMode: FailureMode
+get(key): value
+update(key, value)
reads mode
DecisionEngine
-hostScores: map
+decide(flow, evidence): Verdict
+addHostEvidence(host, weight)
Enforcer
-allowlist: set
-maxBlocksPerMinute: int
+enforce(verdict): BlockEntry
+unblock(ip)
manages
BlockEntry
+ip: string
+reason: string
+duration: int
+offenceCount: int
+createdAt: datetime
produces
acts on
«enumeration» HealthState
NORMAL
DEGRADED
FAILED
RECOVERING
«enumeration» FailureMode
FAIL_OPEN
FAIL_CLOSED
DEGRADED
Verdict
+action: Action
+attackClass: string
+layers: Layer[]
+explanation: string
«enumeration» Action
PASS
ALERT
BLOCK
EventStore
+save(record)
+query(filter): list
+purgeOlderThan(days)
logs
logs